Pressure in a Liquid

Exam code: 0625 & 0972

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Liquid pressure

Factors affecting pressure in a liquid

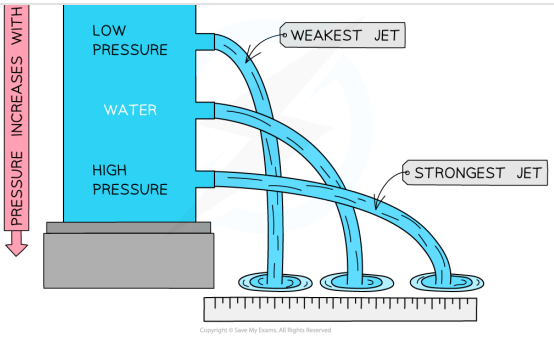
- The **pressure** beneath the surface of a liquid will **increase** with the **depth** of the liquid
  - This is because the pressure in a liquid is caused by the force of the **weight** of the liquid above it
  - As the depth increases, there is increasingly more liquid above which causes the pressure to increase
- Pressure in a liquid also depends on the **density** of the liquid
  - The more dense the liquid, the greater the pressure it exerts

Pressure in a column of water

- In a column of water, the **highest pressure** would be at the **bottom**
  - If a hole is made at the bottom of the column, the water will pour out with a **large** force
  - If a hole was made at the top of the column, the water will pour out with a **small** force
  - This is because of the **difference in pressure** in the column caused by the weight of the water

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Pressure in a column of water increases with depth, shown by the strong and weak jet of water

- If several holes were made at the same height, the water would spurt out the same distance from each hole
  - This is because pressure at the **same** depth acts **equally** in all directions

Examiner Tips and Tricks

Remember, a fluid can be either a **liquid** or a **gas**. In all fluids, pressure is affected by depth and density. This means that, like in liquids, the pressure within gases also changes with depth and density.

Calculating liquid pressure

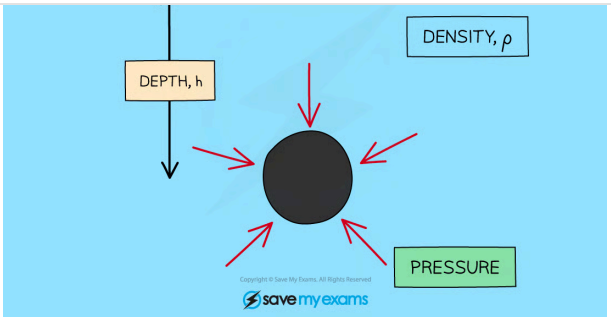
Extended tier only

- The pressure acting on an object in a liquid changes with **depth**
  - The **deeper** the object, the higher the pressure exerted upon it and vice versa
- The equation for the pressure difference at different depths in a liquid is given by the equation:

$$\Delta p = \rho g \Delta h$$

- Where:
  - Δp = change in pressure, measured in pascals (Pa)
    - Where 1 Pa = 1 N/m^2
  - ρ = density of the liquid, measured in kilograms per metre cubed (kg/m^3)
  - g = gravitational field strength on Earth, measured in newtons per kilogram (N/kg)
  - Δh = change in height of the column, measured in metres (m)

Pressure in a liquid with a density is applied at a depth



The force from the pressure of objects in a liquid is exerted evenly across its whole surface

Worked Example

Calculate the depth of water in a swimming pool where a pressure of 20 kPa is exerted. The density of water is 1000 kg/m^3 and the gravitational field strength on Earth is 9.8 N/kg.

Answer:

Step 1: List the known quantities

- Pressure, $\Delta p = 20\ \text{kPa}$
- Density of water, $\rho = 1000\ \text{kg/m}^3$
- Gravitational field strength, $g = 9.8\ \text{N/kg}$

Step 2: List the relevant equation

$$\Delta p = \rho g \Delta h$$

Step 3: Rearrange for height, Δh

$$\Delta h = \frac{\Delta p}{\rho g}$$

Step 4: Convert any units

$$20\ \text{kPa} = 20\,000\ \text{Pa}$$

Step 5: Substitute in the values

$$\Delta h = \frac{20\,000}{1000 \times 9.8}$$

$$\Delta h = 2.0408 = 2.0\ \text{m}$$

Examiner Tips and Tricks

You need to be able to recall the equation for the change in pressure and make sure you are comfortable rearranging it for the variable required in the question!

Some exam questions may ask about the total pressure exerted on an object or surface within the liquid; in this case,

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